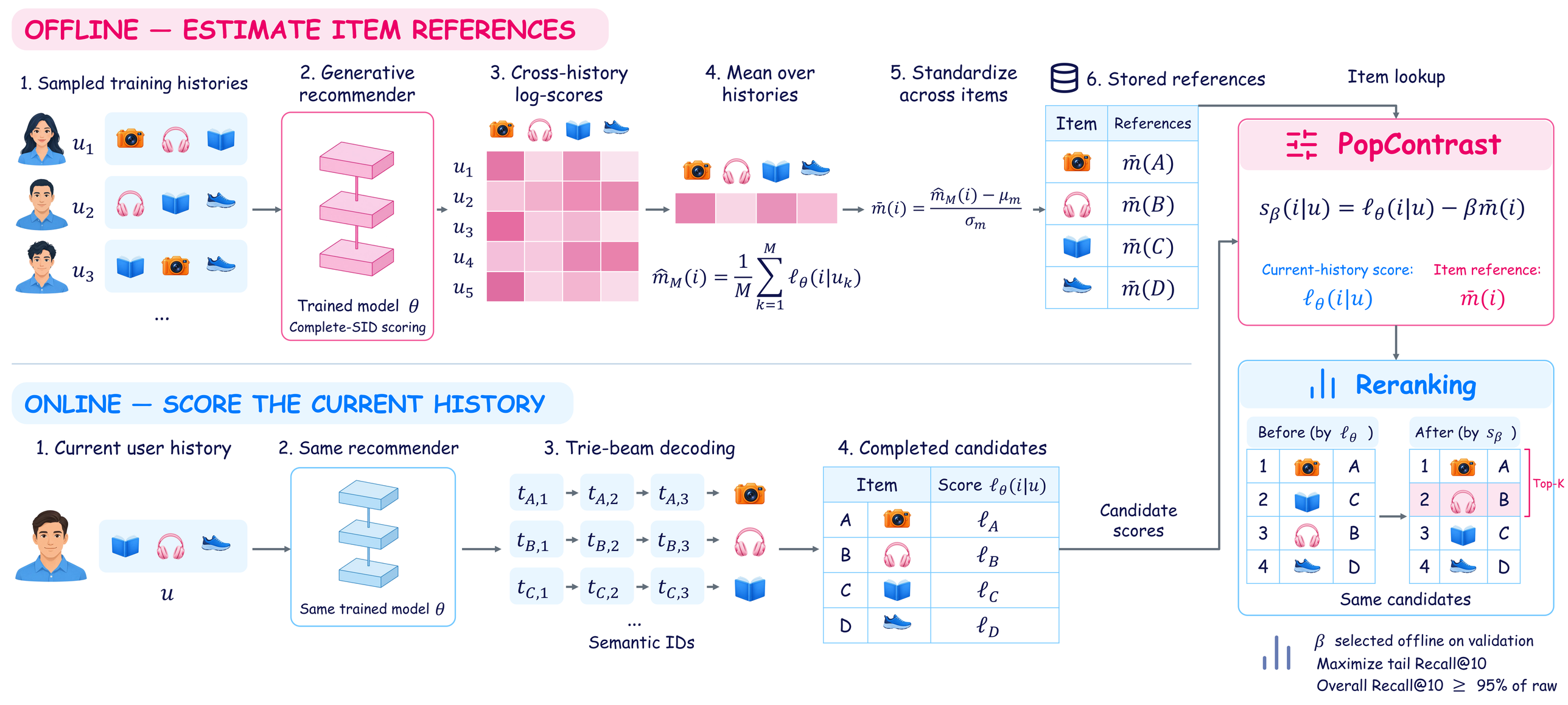

OFFLINE — ESTIMATE ITEM REFERENCES
1. Sampled training histories
u_1
u_2
u_3
...
2. Generative recommender
Trained model θ
Complete-SID scoring
3. Cross-history log-scores
u_1
u_2
u_3
u_4
u_5
4. Mean over histories
$\hat{m}_M(i) = \frac{1}{M}\sum_{k=1}^{M} \ell_\theta(i|u_k)$
5. Standardize across items
$\bar{m}(i) = \frac{\hat{m}_M(i) - \mu_m}{\sigma_m}$
6. Stored references
Item
References
$\bar{m}(A)$
$\bar{m}(B)$
$\bar{m}(C)$
$\bar{m}(D)$
Item lookup
PopContrast
$s_\beta(i|u) = \ell_\theta(i|u) - \beta\bar{m}(i)$
Current-history score:
$\ell_\theta(i|u)$
Item reference:
$\bar{m}(i)$
Reranking
Before (by ℓ_θ)
1 A
2 C
3 B
4 D
After (by s_β)
1 A
2 B
3 C
4 D
Top-K
Same candidates
ONLINE — SCORE THE CURRENT HISTORY
1. Current user history
u
2. Same recommender
Same trained model θ
3. Trie-beam decoding
$t_{A,1} \to t_{A,2} \to t_{A,3}$
$t_{B,1} \to t_{B,2} \to t_{B,3}$
$t_{C,1} \to t_{C,2} \to t_{C,3}$
...
Semantic IDs
4. Completed candidates
Item
Score $\ell_\theta(i|u)$
A ℓ_A
B ℓ_B
C ℓ_C
D ℓ_D
Candidate scores
β selected offline on validation
Maximize tail Recall@10
Overall Recall@10 ≥ 95% of raw